



# Defense Operations Intelligence Platform (DOIP) — Master Project Document

**Document Type:** Complete Project Specification & Implementation Guide

**Version:** 1.7 · **Date:** July 2026 · **Author:** Saranraj U

**Classification:** Educational / Simulation-Only — No IoT, no drones, no real-world military control. 100% software-based.



**Scope Guarantee:** DOIP is a *simulation, training, and decision-support* platform. Every sensor, unit, drone, and incident in the system is synthetically generated by software. The platform never interfaces with physical hardware, weapons, or real-world command systems. This makes it legally safe, ethically clean, and fully buildable by a solo developer or small team.

## How to Read This Document

This master document is organized into **19 chapters**, each a dedicated sub-page below. Together they form the complete specification — vision, justification, requirements, architecture, data design, AI systems, implementation phases, security, deployment, testing, and appendices.

| # | Chapter                                       | What it covers                                                          |
|---|-----------------------------------------------|-------------------------------------------------------------------------|
| 1 | Executive Summary & Vision                    | What DOIP is, why it's needed, goals, users, use cases                  |
| 2 | Product Requirements & Feature Specifications | All 10 core modules in full functional detail                           |
| 3 | System Architecture                           | Diagrams, service breakdown, communication patterns, evolution strategy |

| #  | Chapter                                              | What it covers                                                                                                                          |
|----|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 4  | Technology Stack                                     | Every technology chosen, with justification and alternatives                                                                            |
| 5  | Data Model & Database Design                         | Full PostgreSQL + PostGIS schema, geospatial queries                                                                                    |
| 6  | API & Real-Time Communication Design                 | REST endpoints, WebSocket protocol, event contracts                                                                                     |
| 7  | Simulation Engine Design                             | The heart of the platform — tick engine, scenarios, event generation                                                                    |
| 8  | AI & Machine Learning Components                     | LLM, RAG, computer vision, predictive analytics                                                                                         |
| 9  | Implementation Strategy & Phase Roadmap              | All 7 phases broken into sprints with deliverables and exit criteria                                                                    |
| 10 | Security, Deployment & DevOps                        | Auth, RBAC, Docker, Kubernetes, CI/CD, monitoring                                                                                       |
| 11 | Testing, Risks & Success Metrics                     | Test strategy, risk register, KPIs, demo script                                                                                         |
| 12 | Appendices                                           | Glossary, repository structure, learning resources, FAQ                                                                                 |
| 13 | AI-Generated Scenario Simulation & Animated Playback | Natural-language what-if scenarios, headless fast runs, animated map playback with AI narration                                         |
| 14 | Budget, Capability & Historical Benchmarking Module  | Budget/inventory analysis, capability gap detection, budget-constrained acquisition recommendations, real-world case-study benchmarking |
| 15 | Multi-Operator Collaboration & War-Room Mode         | Shared live sessions, role-based consoles, map annotations, presence, task assignment, team debriefs                                    |
| 16 | Adversarial Red-Team Simulation Agent                | Adaptive AI training adversary: performance-aware complication injects with fairness rules and full debrief transparency                |
| 17 | Project & Platform Workflows                         | End-to-end workflows: master cycle, runtime heartbeat, user journeys, AI pipelines, training exercises, development & data lifecycle    |

| #  | Chapter                                      | What it covers                                                                                                                                |
|----|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 18 | UI/UX Design Specification                   | Design philosophy, free design toolchain, design tokens, screen inventory, wireframes, component library, interaction patterns, accessibility |
| 19 | Starter Content, Licensing & Zero-Cost Stack | Starter scenario library, RAG corpus, prompt library, licensing & legal decisions, project cost sheet, free-tools audit                       |

## Executive Abstract

The **Defense Operations Intelligence Platform (DOIP)** is a software-only command and decision-support platform for **military simulation, training, and defense operations planning**. It presents an operator with a single integrated picture: a live interactive map of simulated units and incidents, real-time event feeds, environmental conditions, logistics status, personnel health, and AI-generated situation summaries and recommendations.

Conceptually, DOIP fuses ideas from Google Maps (geospatial visualization), Grafana (operational dashboards), ChatGPT (AI assistance), flight tracking (live entity movement), and incident-management platforms (structured response workflows) into one coherent application.

The project doubles as a **portfolio-grade demonstration of advanced software engineering**: real-time distributed systems, GIS, event streaming, applied AI (LLM + RAG + computer vision), predictive analytics, and production-style DevOps — all achievable without any physical hardware.

## Guiding Principles

1. **Simulation-first.** The synthetic data generator is the core product; everything else consumes its events.
2. **Working software early.** A modular monolith and Docker Compose first; microservices, Kafka, and Kubernetes are introduced as deliberate, documented migrations in later phases.
3. **Every feature must demo.** Each phase ends with something an observer can watch happen live on the map.
4. **AI as an assistant, not an authority.** The AI summarizes, retrieves, and recommends — the human operator always decides.

5. **Ethics by design.** No real weapons data, no targeting functionality, no hardware control. Scenario content is modeled on defensive and support operations — patrol, surveillance, convoy escort, casualty evacuation, logistics, and military HADR deployments — never offensive strike planning.
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*The chapters below contain the full document. Read them in order for the complete picture, or jump to the chapter you need.*

-  [Chapter 1 — Executive Summary & Vision](#)
-  [Chapter 2 — Product Requirements & Feature Specifications](#)
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